Speaker script

"Back to entropy": choosing a base model by how well it reads your text

Deck `slides/talk.pdf`: 11 slides + 4 backup slides
Target about 7 minutes with pauses (600 words), then Q&A
Goal The team leaves with one idea: to choose a base model, score it on the domain's own documents

How to use this document. Italic text is what to say, written to be spoken, not read verbatim. Orange brackets are delivery cues. Times are cumulative. The spoken length is checked by `python tools/script_timing.py`, which fails the build if it passes ten minutes.

The spine: if only this lands, the talk worked:

1. Bits per byte on whole documents of the domain's text, before any training, anticipates which base model fine-tunes best; GSM8K and MMLU-Pro do not.
2. Two details decide it: the text has to be the task's, and each document has to be read whole from its start.
3. No labels, no task and no training are needed.

SLIDE 1 0:00–0:15

Back to entropy

This talk is about one question: which base model will fine-tune best on our data, and how to know before we fine-tune anything.

SLIDE 2 0:15–1:00

Which base model will fine-tune best on our data?

Fine-tuning every candidate would settle it, but that needs a task, labels and a training run per model. So we read a scorecard instead: MMLU-Pro, GSM8K, HellaSwag. Those grade a model before the fine-tuning it will get. Look at these two small models. Two points apart on HellaSwag, two percent apart on our news text, fifty-five points apart on GSM8K. Fine-tuned on news, they cannot be told apart.

SLIDE 3 1:00–1:55

Read the domain's documents; the fine-tuned order follows

Here is the result. Fifteen base models, each fine-tuned on the same news-writing task. Each line joins a model's rank before fine-tuning to its rank after. On GSM8K and MMLU-Pro the lines cross everywhere. On the right we rank by bits per byte on whole news documents, with no training at all, and the lines run almost flat: every model lands within one rank of where it finishes. The blue line is Gemma-4-12B, ninth on GSM8K, and it builds the best fine-tuned system of the fifteen.

[LAND THIS: one forward pass, and the fine-tuned order is already there.]

SLIDE 4 1:55–2:40

Pair by pair

The fair test is pairs of models of similar size, because between very different sizes the bigger one usually wins. There are thirty-nine such pairs. Document bits per byte orders thirty-seven of them the way the fine-tune does. HellaSwag thirty-four, size twenty-eight, and GSM8K and MMLU-Pro are a coin flip. So is each model's own score on the task before any training: fine-tuning reorders the models, and the document reading predicts the new order. A second, very different fine-tuning recipe gives the same picture.

SLIDE 5 2:40–3:25

Why this is not already how it is done

If this sounds obvious, the literature says the opposite. The closest test we know of fine-tuned fifty one-billion models and found that perplexity ordered them worse than chance. Two details explain the difference, and the usual practice gets both wrong: which text you score, and how you cut it.

SLIDE 6 3:25–4:15

Detail 1: blocks cut from a stream misrank strong models

The common way to compute perplexity joins the test set into one stream and cuts it into blocks. Most blocks then start in the middle of a document, with no start token. Some models pay heavily for that. Gemma opens each block far worse than a model eight times smaller and has passed it by the end. On blocks Gemma comes thirteenth of fifteen; on whole documents, second; fine-tuned, first.

SLIDE 7 4:15–5:05

Detail 2: the text has to be like the task's

Second, the text. Score the same models on arXiv mathematics and use that to predict the news fine-tune, and you get twenty-three of thirty-nine pairs, worse than ranking by size. And which benchmark the ranking agrees with follows the text: on news, Reddit and Hacker News it is HellaSwag; on mathematics it is GSM8K and MMLU-Pro. We wrote that rule down before reading the new corpora.

SLIDE 8 5:05–5:50

No training, and no noise

We started by adapting every model briefly to the domain first, and on blocks that is a repair. On whole documents it lowers every model's loss but barely changes the order: one pair in thirty-nine. So no training is needed. And the reading is deterministic: run it again and you get the same number to the last digit, where the same weights on a new machine move GSM8K by more than half a point.

SLIDE 9 5:50–6:30

Why loss is the right quantity

Why should this work? A base model learns everything it knows by minimising this loss, and fine-tuning minimises it again on the task. The loss splits into the entropy of the text, the same for every model, plus the model's distance from the text. Per byte, every tokenizer is on one scale, and on fresh text no model has trained on it.

SLIDE 10 6:30–7:00

Using it, and adding your model

The pipeline reports this number for every candidate in one forward pass. The leaderboard now ranks by it, on news and on a public maths corpus anyone can run.

SLIDE 11 7:00–7:08

Score the candidates on your own documents

So: score the candidates on your own documents. Thank you, happy to take questions.

Q&A preparation

“Why not just fine-tune on the task?” When you have the task, its labels and the budget, do. A base model is usually chosen before any of those exist.

“Does it beat HellaSwag?” It has the top point estimate, 37 against 34 pairs, but with fifteen models the intervals overlap. HellaSwag is close on news; on mathematics text the ranking follows GSM8K and MMLU-Pro instead, so HellaSwag is not a general substitute.

“Only one task?” Yes, one fine-tuned task. We tried two on mathematics; on both a 0.5B and a 12B model finished within 0.012 ROUGE-L, so the tasks could not rank models and we stopped them by a rule fixed in advance.

“What about contamination of our own text?” Text dated after every release, deduplicated with no split overlap, and membership screens with a quarantine of anything any model flags.

“What does it cost?” One forward pass over about six megabytes of text per model.